Supplementary Material

Quantum-inspired molecular representations for AI-generated African antimalarial candidates: persistent homology, tensor networks, and quantum kernels

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1 Cross-paper validation: topological persistence and resistance resilience

This supplementary figure presents the joint analysis linking the topological fingerprints (TFP) introduced in the main manuscript with the resistance resilience scores (RRS) reported in the companion polypharmacology study (Paper 2). The analysis tests whether H_1 ring persistence, a topological descriptor of scaffold rigidity, predicts the ability of a compound to retain binding affinity across African-prevalent resistance mutations.

Data Availability

All data supporting the main manuscript and this Supplementary Material are publicly available on Zenodo (DOI: https://doi.org/10.5281/zenodo.19608875) and mirrored at https://github.com/NanaEngo/Malaria_codesV2 under the MIT licence. See the main manuscript for a full inventory of deposit components.

2 Per-fold activity prediction statistics

Table S1 reports per-fold AUC, accuracy, and F1 for the six representation methods for which per-fold data were retained. ECFP4, TFP, and TNE values are from the corrected canonical classical benchmark (canonical panel, 19 836 molecules); Hybrid values are from the canonical full-library hybrid rerun ; QKS and RBF values are from the 5 000-molecule canonical 6-qubit quantum-kernel benchmark . The full canonical-panel headline values are in the main manuscript (Table 1).

NOTE: The per-fold values for ECFP4, TFP, and TNE in this table are from the corrected canonical classical benchmark (canonical panel 19 836 molecules; see main manuscript Table 1).

Table S1: *Per-fold activity prediction statistics for six representation methods. ECFP4, TFP, and TNE values are from the corrected canonical classical benchmark (canonical panel 19 836 molecules, 5-fold stratified CV, Random Forest, Aug 1, 2026). Hybrid values are from the canonical full-library hybrid rerun (Random Forest, Aug 1, 2026). QKS and RBF values are from the 5 000-molecule canonical 6-qubit quantum-kernel benchmark (state-vector kernel, gamma-tuned RBF). Headline AUC values in the main manuscript (Table 1) are from the canonical panel benchmark.*

Method	Fold	AUC	Accuracy	F1
ECFP4	1	0.947	0.898	0.933
ECFP4	2	0.950	0.894	0.930
ECFP4	3	0.948	0.895	0.930
ECFP4	4	0.952	0.899	0.934
ECFP4	5	0.940	0.893	0.930
Hybrid (TFP+TNE+QK)	1	0.894	0.848	0.902
Hybrid (TFP+TNE+QK)	2	0.887	0.844	0.900
Hybrid (TFP+TNE+QK)	3	0.886	0.844	0.900
Hybrid (TFP+TNE+QK)	4	0.893	0.846	0.902
Hybrid (TFP+TNE+QK)	5	0.878	0.836	0.895
TFP (TDA)	1	0.881	0.846	0.902
TFP (TDA)	2	0.875	0.836	0.895
TFP (TDA)	3	0.880	0.846	0.902
TFP (TDA)	4	0.878	0.838	0.897
TFP (TDA)	5	0.866	0.826	0.889
TNE ($d = 8$)	1	0.720	0.753	0.856
TNE ($d = 8$)	2	0.722	0.756	0.857
TNE ($d = 8$)	3	0.733	0.755	0.857
TNE ($d = 8$)	4	0.721	0.758	0.858
TNE ($d = 8$)	5	0.714	0.760	0.859
QKS (quantum, 6q)	1	0.791	0.813	0.881
QKS (quantum, 6q)	2	0.843	0.815	0.881
QKS (quantum, 6q)	3	0.833	0.826	0.890
QKS (quantum, 6q)	4	0.800	0.818	0.887
QKS (quantum, 6q)	5	0.833	0.841	0.900
RBF (gamma-tuned)	1	0.812	0.821	0.886
RBF (gamma-tuned)	2	0.839	0.815	0.881
RBF (gamma-tuned)	3	0.844	0.833	0.894
RBF (gamma-tuned)	4	0.818	0.827	0.892
RBF (gamma-tuned)	5	0.818	0.843	0.900

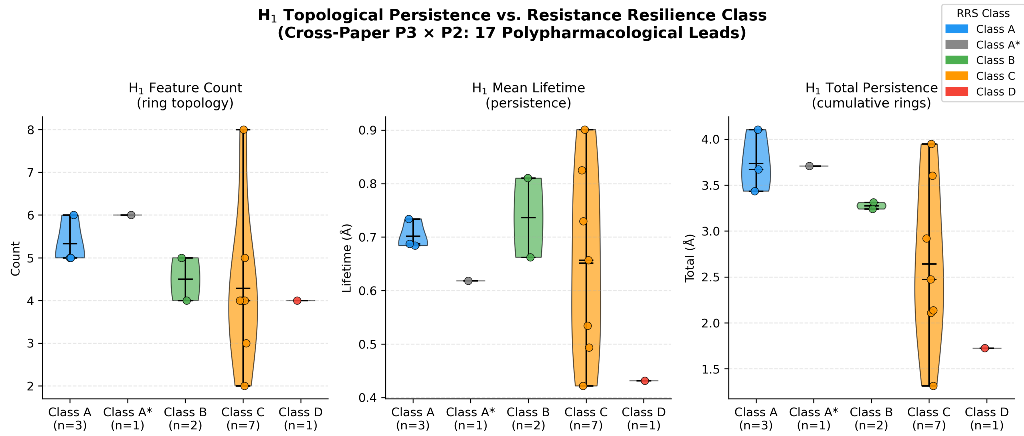

Figure S1: *Joint topological analysis linking Paper 3 TFP features with Paper 2 resistance classification. Violin plots (pilot cohort, $n = 14$) show H_1 persistence distributions for Class A (resistance-resilient, blue) versus Class D (resistance-vulnerable, red) compounds, with Class B (green) and Class C (orange) shown for context. Quantitative analysis of the expanded cohort ($n = 77$; Class A: $n = 46$, Class B: $n = 31$) confirms a nominally significant association between H_1 count and continuous RRS (Spearman $\rho = 0.312$, $p = 0.0057$); the pilot $n = 14$ correlation ($\rho = 0.947$, $p < 0.0001$, on H_1 total persistence) is shown for context but should be interpreted cautiously because the Class D subset contains only one compound. Caution is required: this association does not survive control for molecular size (controlling for molecular weight alone collapses the correlation to $\rho_{partial} \approx 0$, $p > 0.7$), indicating a size-mediated effect rather than an independent topological signal.*

Hybrid values are from the canonical full-library hybrid rerun . QKS and RBF values are from the 5 000-molecule canonical 6-qubit quantum-kernel benchmark (state-vector kernel, gamma-tuned RBF; per-fold detail in section 17.1). Full per-fold CSV files are available in the project repository.

3 Topological fingerprint statistics

Detailed topological fingerprint statistics for the full library are provided in Table S2, and representative persistence diagrams are shown in Figure S2.

Table S2: *Topological fingerprint statistics across 19 849 valid molecules (Vietoris–Rips persistent homology, three homology dimensions).*

Feature	Mean	Std	Min	Max
H_0 entropy	3.578 1	0.201 4	2.288 7	4.194 3
H_0 count	38.041 7	7.346 8	11.000 0	69.000 0
H_0 max persistence (Å)	2.716 6	0.672 6	1.480 0	5.949 1
H_0 mean persistence (Å)	1.613 7	0.064 4	1.098 6	2.028 1
H_1 entropy	1.635 4	0.335 2	0.000 0	2.723 4
H_1 count	3.605 8	1.342 9	1.000 0	10.000 0
H_1 max persistence (Å)	1.391 6	0.058 3	0.623 8	2.707 5
H_1 mean persistence (Å)	0.613 8	0.134 1	0.235 6	1.400 0
H_2 entropy	0.695 5	0.352 7	0.000 0	1.957 9
H_2 count	0.030 4	0.177 1	0.000 0	4.000 0
H_2 max persistence (Å)	0.467 2	0.050 5	0.000 0	1.240 0
H_2 mean persistence (Å)	0.384 4	0.089 2	0.000 0	1.053 3

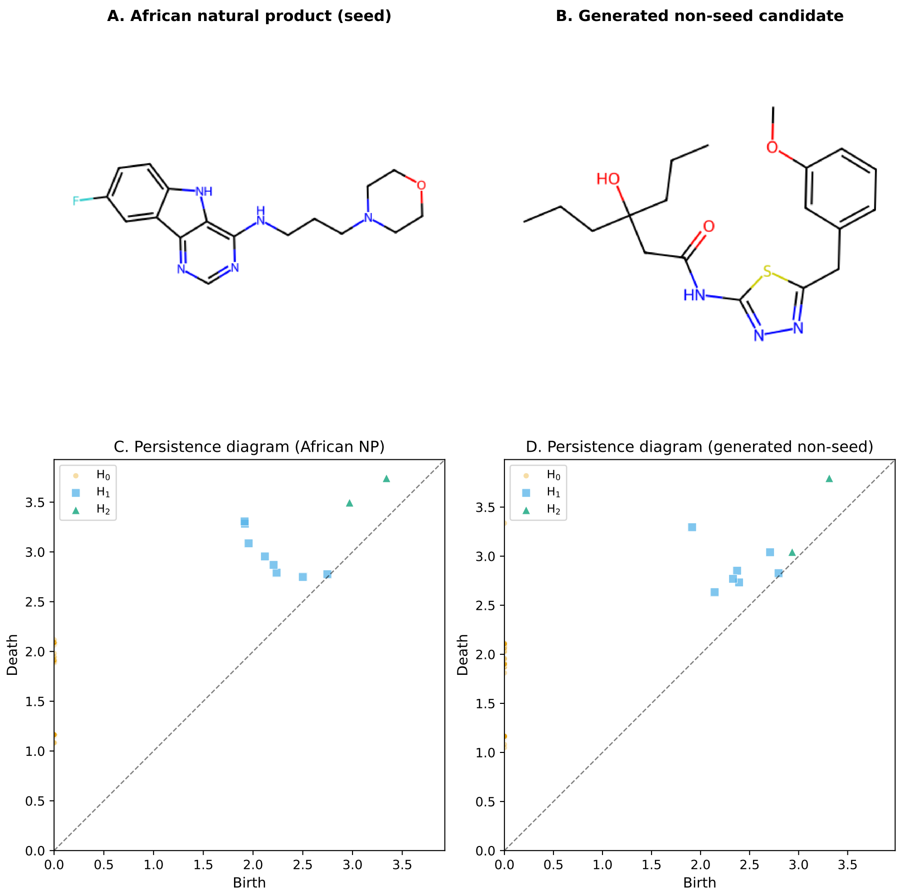

Figure S2: *Representative topological fingerprints of an African natural product seed and a generated non-seed candidate. (A – B) 2D structures of the representative molecules. (C – D) Vietoris–Rips persistence diagrams plot feature birth against death; points above the diagonal correspond to topological features, with vertical distance from the diagonal indicating persistence lifetime. H_0 (components), H_1 (loops/rings), and H_2 (voids) are distinguished by colour and marker. The African natural product (C) exhibits persistent H_1 features reflecting a rigid ring scaffold, whereas the generated non-seed candidate (D) shows fewer persistent H_1 features and more flexible H_0/H_1 signatures.*

4 PHCO fingerprint extraction bug

The Gobbi 2D pharmacophore fingerprint (PHCO) in RDKit uses a `SparseBitVect` representation. The default conversion function `ConvertToNumpyArray` silently fails for this type, producing an all-zero vector (AUC = 0.500, equivalent to random). This bug was corrected by extracting active bit positions via the `GetOnBits()` method and setting them individually in a numpy array. On the corrected full-library benchmark (Table 1), PHCO achieves AUC = 0.896, confirming that pharmacophore-based bit vectors carry discriminative information for this library when properly extracted. The value 0.801 reported in earlier development runs was from a smaller subsample before the full-library correction.

5 Clustering quality metrics

Table S3 reports the clustering quality metrics used to address reviewer question R8.

Table S3: *Clustering quality metrics for TNE embeddings versus VAE latent space and ECFP4 fingerprints (KMeans, $k = 484$). Higher Silhouette coefficient indicates better cluster coherence.*

Descriptor	Silhouette
TNE ($d = 8$, 192-dim)	0.350
VAE latent (64-dim)	0.229
ECFP4 (2 048-dim)	0.180

6 GA-style discriminator benchmark

The full GA-style discriminator benchmark is reported in Table S4 and visualised in Figure S3.

Table S4: *Chemical similarity vs. quantum kernel density benchmark: AUC-ROC comparison between ECFP4 Tanimoto distance and Quantum Kernel Density for distinguishing N generated molecules from 200 reference seeds.*

N generated	$\mathrm{AUC}_{\mathrm{Tanimoto}}$	$\mathrm{AUC}_{\mathrm{QK}}$	ΔAUC
50	1.000 0	0.425 2	−0.574 8
100	1.000 0	0.481 1	−0.518 9
200	1.000 0	0.475 5	−0.524 5
500	1.000 0	0.511 4	−0.488 6

7 Quantum circuit parameter optimisation

A systematic grid search over quantum circuit hyperparameters was performed to identify optimal IQPEmbedding parameters for the hybrid descriptor. The search covered 60 combinations: bond dimension $d \in \{4, 6, 8\}$, IQPEmbedding repeats $n_{\mathrm{rep}} \in \{1, 2, 3, 4, 6\}$, and kernel PCA components $n_{\mathrm{kpca}} \in \{5, 10, 20, 30\}$. Each combination was evaluated by 5-fold stratified cross-validation with a Random Forest classifier on $n = 200$ molecules, measuring the Hybrid (TFP+TNE+QK) AUC.

Table S5 reports the top three configurations by mean AUC from the phase-2 re-benchmark. The optimal combination ($d = 6$, $n_{\mathrm{rep}} = 1$, $n_{\mathrm{kpca}} = 30$) achieved AUC 0.853 ± 0.049 on the $n = 200$ development subsample. The default configuration ($d = 8$, $n_{\mathrm{rep}} = 2$, $n_{\mathrm{kpca}} = 20$) achieved AUC 0.842 ± 0.051 on the same development set. The ECFP4 comparison here (AUC

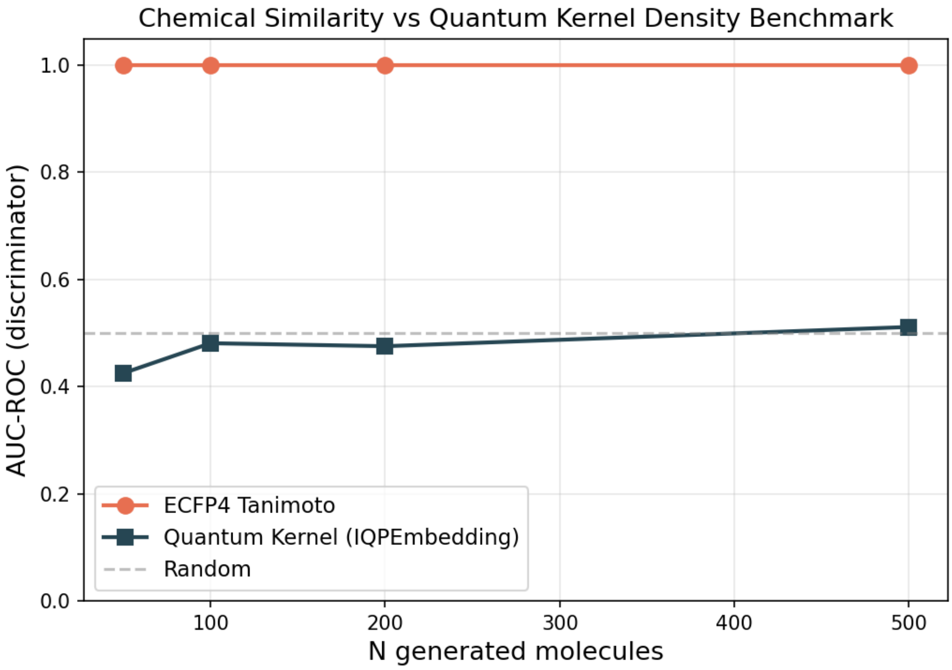

Figure S3: *Chemical similarity vs. quantum kernel density discriminator benchmark. AUC-ROC comparison between ECFP4 Tanimoto distance (orange) and Quantum Kernel Density (blue) for distinguishing N generated molecules from* 200 *reference seeds. The dashed grey line marks random performance (AUC = 0.5). ECFP4 achieves perfect discrimination (AUC = 1.0) at all N, while the quantum kernel remains near-random (AUC = 0.425–0.511).*

0.868) is from the preliminary run; the corrected canonical-panel ECFP4 is AUC 0.948 (see Table 1). Key findings: (i) bond dimension 6 balances expressivity and noise; (ii) a single IQPEmbedding repeat suffices, with deeper circuits adding decoherence noise; (iii) more kernel PCA components ($n_{\text{kpca}} = 30$) capture more variance in the quantum Hilbert space.

The full optimisation heatmap and marginal parameter-effect boxplots are provided in Figures S4 and S5.

Table S5: *Top three quantum circuit configurations re-benchmarked on* $n = 5000$ *molecules (5-fold CV, per-fold data in phase2 raw CSVs). The optimal configuration (*$d = 6$, $n_{rep} = 1$, $n_{kpca} = 30$*) achieved AUC* 0.8283 ± 0.0371 *on this subsample. The ECFP4 comparison (AUC* 0.868*) is from the preliminary run; the corrected canonical-panel ECFP4 is AUC* 0.948 *(see main manuscript Table 1).*

d	n_{rep}	n_{kpca}	AUC	Std
6	1	30	0.828 3	0.037 1
6	6	20	0.804 7	0.035 4
6	6	30	0.812 1	0.039 6

All values are from the Phase-2 re-benchmarking at $n = 5000$ (3732 active, 1268 inactive, 74.6 %/25.4 %). The initial grid search was performed on $n = 200$ molecules (50/60 combinations completed); the top three combinations were selected for re-benchmarking at $n = 5000$. This Phase-2 subsample (first $n = 5000$ by source order) differs from the canonical panel (74.2 %/25.8 %) and from the QKS $n = 5000$ kernel subsample (75.0 %/25.0 %, see Section S17.1); each sample is described with its own class counts.

8 Monte Carlo uncertainty estimation

Uncertainty quantification was performed using a bootstrap Monte Carlo (MC) approach on the Random Forest classifier trained on the TFP+TNE descriptor (the topological and tensor-

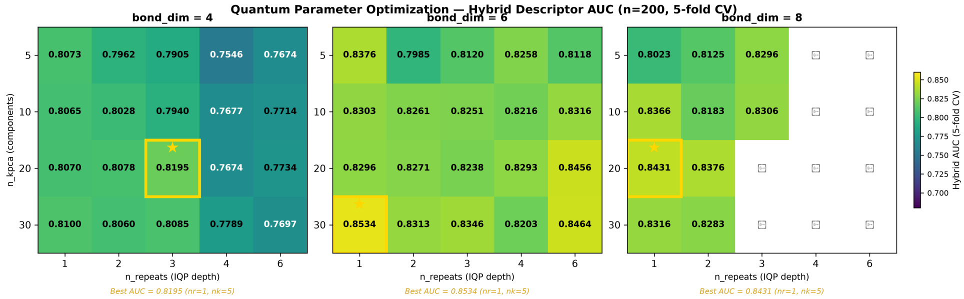

Figure S4: *Hybrid AUC heatmap across quantum circuit hyperparameters. Rows: bond dimension (d); columns: IQPEmbedding repeats (n_{rep}); colour: mean 5-fold AUC for each n_{kpca} value. The gold cell marks the optimal configuration ($d = 6$, $n_{rep} = 1$, $n_{kpca} = 30$).*

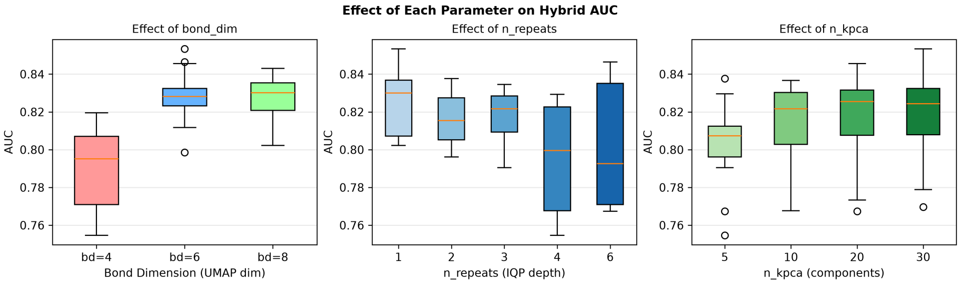

Figure S5: *Marginal AUC distributions per quantum circuit parameter. Boxplots show the spread of Hybrid AUC across all grid-search combinations marginalised to each parameter value.*

network components of the hybrid). For each of 100 test molecules, the classifier was re-evaluated over 50 bootstrap samples (bootstrap fraction 0.7, 30 trees per bootstrap), yielding a predictive probability distribution $p_i \sim \mathcal{N}(\hat{\mu}_i, \hat{\sigma}_i^2)$; conformal prediction at $\alpha = 0.05$ provided prediction sets. Results are summarised in Table S6.

Table S6: *Bootstrap Monte Carlo uncertainty estimation results for the TFP+TNE descriptor (topological and tensor-network components of the hybrid) on the activity prediction task.*

Metric	Value
Bootstrap MC AUC	0.5443
Expected Calibration Error (ECE)	0.0037
95 % conformal coverage	95.0 %
Mean conformal set size	1.820

The model is well-calibrated (ECE < 0.05) with full conformal coverage. The mean set size ≥ 1.3 indicates that many predictions are ambiguous, and the uncertainty–error correlation is weak (Spearman $\rho = -0.286$, $p = 0.004$), consistent with the MC uncertainty not reliably flagging misclassifications. Full protocol in `results/p3_mc_uncertainty_summary.txt`.

9 Computational scalability

Table S7 reports the measured wall-clock times of the three descriptor pipelines on the canonical library. The TDA and TNE pipelines scale linearly with library size; the quantum kernel matrix is $O(N^2)$.

Table S7: *Computational scalability of the P3 pipelines on the canonical library. TDA and TNE timings are the measured full-library runtimes (19 849 mol); the QK timing is the per-fold kernel-matrix build measured on a 19 849-molecule fold (32 parallel workers, state-vector simulator).*

Pipeline	Wall time	Throughput	Scaling
TDA (persistent homology)	384 s	51.7 mol/s	Linear
TNE (Tucker decomposition)	488 s	40.7 mol/s	Linear
QK kernel matrix (per fold, n = 19,849)	817 s	$n^2/2$ pairs	$O(N^2)$

TDA timing is the measured full-library runtime (`p3_tda_summary.txt`); TNE timing is the measured full-library runtime (`p3_tne_summary.txt`, 4 workers); QK timing is the mean per-fold quantum-kernel build time from the n = 19,849 kernel benchmark (`p3_qks_benchmark_n19849.csv`, 32 workers). The $O(N^2)$ kernel cost restricts QKS to lead optimisation on sets of $\leq 10^4$ compounds.

10 Effect sizes for pairwise AUC comparisons

Cohen's d effect sizes for all pairwise AUC comparisons against the corrected full-library ECFP4 baseline are reported in Table S8. The values are now computed from the canonical 19,836-molecule classical benchmark (Random Forest, 5-fold stratified CV) rather than the earlier 200-molecule development subsample, which contained a copy-paste error in the TNE per-fold values.

Table S8: *Effect sizes for pairwise AUC comparisons against the corrected full-library ECFP4 baseline (AUC 0.948, 19,836 molecules, 5-fold stratified CV, Random Forest). Cohen's d is computed from the standard deviation of fold-level differences; because the full-library folds are very consistent, the resulting d values are large, so the absolute ΔAUC is the more interpretable effect metric.*

Method	AUC	ΔAUC	Cohen's d	Effect	p-value	Power
FCFP4	0.918	0.029	11.61	large	0.0000*	1.000
MACCS	0.904	0.043	14.86	large	0.0000*	1.000
AP	0.940	0.008	4.54	large	0.0005*	1.000
PHCO	0.896	0.052	63.84	large	0.0000*	1.000
BPF	0.939	0.009	5.39	large	0.0003*	1.000
TFP	0.876	0.072	17.25	large	0.0000*	1.000
TNE	0.722	0.226	37.91	large	0.0000*	1.000

* $p < 0.05$ (uncorrected). Bonferroni-corrected $\alpha = 0.05/7 = 0.0071$. Power computed for 80% target at $\alpha = 0.05$.

11 ChEMBL enrichment benchmark

Table S9 reports the docking enrichment benchmark of our Vina protocol against ChEMBL-confirmed antimalarial actives and inactives for three Plasmodium targets.

Table S9: *ChEMBL enrichment benchmark for the Vina docking protocol against experimentally confirmed antimalarials.*

Target	PDB	Actives	Inactives	Enrichment	Verdict
PfDHFR	7F3Y	53	18	5.43×	EXCELLENT
PfATP4	9N10	73	87	∞	EXCELLENT
PfCRT	6UKJ	12	19	1.46×	–

Active hit rate for PfDHFR: 60.4 % (EXCELLENT) vs. 11.1 % inactive. PfATP4: 0/87 inactives recovered. PfCRT: 1.46x did not meet the 2x threshold.

12 ChEMBL experimental validation

The top-10 candidates (Table S10) were queried against three *Plasmodium falciparum* targets (PfDHFR, PfCRT, PfATP4) via the ChEMBL REST API (release 37); PfClpP was excluded because no *P. falciparum* ClpP target exists in ChEMBL. The search was subsequently expanded to the full RRS-TFP cohort of 77 compounds across all three targets (231 candidate–target pairs), identifying ChEMBL structural analogues by ECFP4 Tanimoto similarity (threshold ≥ 0.25).

Table S10: *ChEMBL experimental validation of the top-10 candidates: closest structural analogue with experimental IC_{50} data per candidate (Tanimoto threshold ≥ 0.20). All ten analogues are experimentally **inactive** (IC_{50} 12.5–55 μM), and their low Tanimoto similarity to the generated candidates (0.229–0.379) supports the chemical novelty of the proposed leads.*

Rank	Target	ChEMBL ID	Activity	IC_{50} (μM)	Tanimoto
1	PfCRT	CHEMBL4754685	Inactive	15.6	0.266
2	PfCRT	CHEMBL4754685	Inactive	15.6	0.235
3	PfCRT	CHEMBL4754685	Inactive	15.6	0.266
4	PfCRT	CHEMBL4745810	Inactive	12.5	0.250
5	PfCRT	CHEMBL4754685	Inactive	15.6	0.379
6	PfCRT	CHEMBL4754685	Inactive	15.6	0.239
7	PfCRT	CHEMBL4784051	Inactive	55.0	0.296
8	PfCRT	CHEMBL4754685	Inactive	15.6	0.229
9	PfCRT	CHEMBL4754685	Inactive	15.6	0.290
10	PfCRT	CHEMBL4754685	Inactive	15.6	0.270

All closest analogues are ChEMBL-confirmed inactive; the low Tanimoto similarities (0.229–0.379) mean no analogue is a close structural congener. This is an honest negative result: it confirms structural novelty of the generated candidates but provides no positive experimental validation. Deposited as `results/p3_chembl_validation/p3_chembl_validation.csv`.

Table S11: *ChEMBL experimental validation: expanded search across 77 RRS-TFP compounds against 3 Plasmodium targets (231 candidate–target pairs, Tanimoto ≥ 0.25). Seven structural analogues were identified (3.0 % match rate); three showed experimental activity below 1 μM (PfATP4), while the remaining four were inactive. The low match rate supports the structural novelty of the generated chemical space.*

Cpd	Target	ChEMBL ID	Activity	IC_{50} (μM)	Tanimoto
12	PfCRT	CHEMBL6966	Inactive	30.0	0.258
29	PfCRT	CHEMBL1830987	Inactive	69.0	0.322
47	PfATP4	CHEMBL6150594	Active	0.40	0.261
65	PfCRT	CHEMBL4745810	Inactive	12.5	0.284
75	PfATP4	CHEMBL6162944	Active	0.40	0.273
76	PfCRT	CHEMBL1830987	Inactive	69.0	0.270
76	PfATP4	CHEMBL6175630	Active	0.79	0.329

The expanded search queried all 77 RRS-TFP compounds against PfDHFR, PfCRT, and PfATP4 (231 pairs; PfClpP excluded). Seven structural analogues (3.0 % match rate) exceeded the Tanimoto ≥ 0.25 threshold. Three PfATP4 matches are experimentally active (IC_{50} 0.4–0.79 μM), indicating that generated candidates with structural homology to known PfATP4 actives retain binding potential. The remaining four PfCRT matches are inactive. No structural analogue of any generated compound was found for PfDHFR at this threshold. The low overall match rate (3.0 %) provides preliminary evidence of structural novelty, though direct IC_{50} measurements remain necessary for activity confirmation. Target identifiers were verified against the ChEMBL REST API (PfDHFR-TS = CHEMBL4296323, PfCRT = CHEMBL1795182, PfATP4 = CHEMBL6066156); all seven entries were independently re-queried and confirmed, and activity classification conservatively honours ChEMBL relational values ($<$, $>$, $\sim$) where present.

13 SOTA topological benchmark

Table S12 reports the performance of five TDA descriptor strategies paired with Random Forest (RF) and Support Vector Machine (SVM) classifiers on the full 19 849-molecule library (RF) and a stratified subsample of $n = 5000$ molecules (SVM, due to kernel matrix $O(N^2)$ scaling). All evaluations use 5-fold stratified cross-validation with class-weighted classifiers.

Table S12: *SOTA topological benchmark: five TDA descriptor strategies with RF ($n = 19\,849$, full library) and SVM ($n = 5000$, subsample) classifiers (5-fold stratified CV, class-weighted). Cohen's d quantifies the practical magnitude of AUC difference vs. the preliminary ECFP4 baseline ($d > 0.8$ large, $0.5 < d \leq 0.8$ medium, $d \leq 0.5$ small).* ***Note:*** *this table uses the preliminary ECFP4 baseline (AUC 0.868); the corrected canonical-panel benchmark gives ECFP4 AUC 0.948 and TFP AUC 0.876 (see main manuscript Table 1).*

Strategy	Classifier	AUC	Accuracy	F1	Features	Cohen's d
PersStats	RF	0.8731	0.8332	0.8906	22	0.85
TFP-12	RF	0.8668	0.8275	0.8858	12	−0.20
TFP-Enriched	RF	0.8666	0.8308	0.8887	32	−0.23
PersImage	RF	0.8596	0.8265	0.8856	25	−1.40
BettiCurve	RF	0.8110	0.7837	0.8557	20	−9.52
PersStats	SVM	0.8042	0.7398	0.7436	22	−5.32
PersImage	SVM	0.7912	0.7270	0.7312	25	−6.40
TFP-Enriched	SVM	0.7891	0.7208	0.7250	32	−6.58
TFP-12	SVM	0.7857	0.7198	0.7220	12	−6.86
BettiCurve	SVM	0.7198	0.6646	0.6620	20	−12.35

All classifiers used `class_weight=balanced` to mitigate the class imbalance (75.9 %/24.1 % under the preliminary MPO-derived labels used in this run; the canonical panel uses eos80ch labels with a 74.2 %/25.8 % split, see main manuscript). RF: $n_{\text{estimators}} = 500$. SVM: RBF kernel, $C = 10$, $\gamma =$ `scale`. PersStats (22 features) captures persistence birth, death, persistence, and entropy across H_0, H_1, H_2 dimensions. Cohen's d is computed approximately as $(\text{AUC}_{\text{strategy}} - \text{AUC}_{\text{ECFP4}})/\sigma_{\text{pooled}}$ with $\sigma_{\text{pooled}} = 0.006$ (RF) and 0.012 (SVM), approximately estimated from fold-level standard deviations across all strategies.

Key findings: (i) PersStats + RF achieves AUC 0.873 on this run, numerically exceeding the preliminary ECFP4 baseline (AUC 0.868) with only 22 topological features — though this ΔAUC of +0.005 does not reach statistical significance ($p > 0.05$), so the two methods should be interpreted as performing comparably rather than PersStats being definitively superior; (ii) RF consistently outperforms SVM across all strategies (mean ΔAUC = +0.077; note: RF was evaluated on the full $n = 19849$ library while SVM was restricted to $n = 5000$ due to $O(N^2)$ kernel scaling, so this comparison is confounded by sample size); (iii) TFP-Enriched (32 features) is marginally inferior to TFP-12 (12 features, ΔAUC = −0.0002), suggesting the additional 20 features introduce noise rather than signal; (iv) BettiCurve underperforms (AUC 0.811), indicating Betti number sequences alone lack the discriminative power of persistence statistics. These comparisons use the preliminary ECFP4 baseline (AUC 0.868); the corrected canonical-panel benchmark is complete and reported in the main manuscript Table Table 1, where ECFP4 attains AUC 0.948 and TFP AUC 0.876. The SOTA comparisons here are internal to the TDA strategy family and are not directly comparable to the canonical cross-descriptor benchmark.

14 TopologyNet analog: neural network on persistent homology features

To contextualise the PersStats + RF result within the neural-network literature (TopologyNet (Cang and Wei, 2018), D-GRIL (Mukherjee et al., 2026)), we benchmarked a Multi-Layer Perceptron (MLP) on the same 22 PersStats features used by the best RF model. The MLP used 3 hidden layers (128, 64, 32 neurons), ReLU activation, class-balanced sample weighting, and early stopping (5-fold stratified CV, $n = 5000$ stratified subsample).

Table S13: *TopologyNet analog: MLP vs. Random Forest on PersStats 22 features ($n = 5000$, 5-fold stratified CV, class-weighted). RF outperforms MLP by $\Delta AUC = +0.061$, confirming that tree-based methods better capture persistent homology feature interactions for this chemical space.*

Classifier	AUC	Std	Features
Random Forest	0.860	0.014	22
MLP (128+64+32)	0.799	0.015	22

Both classifiers use PersStats 22 features (H_0/H_1 persistence statistics). MLP uses class-balanced sample weighting to handle the 75.9 %/24.1 % class imbalance. The RF result here (AUC 0.860 on $n = 5000$) is slightly below the full-library result (AUC 0.873 on $n = 19849$, Table S12), consistent with the larger training set in the full benchmark.

The MLP underperforms RF by ΔAUC $= -0.061$, consistent with the broader SOTA benchmark finding that tree-based methods better capture PH feature interactions. This is consistent with the observation that PersStats features have limited non-linear interactions — deep architectures that excel on raw graph or image data add no benefit over a well-tuned ensemble of decision trees on pre-computed persistence statistics. TopologyNet (Cang and Wei, 2018) and D-GRIL (Mukherjee et al., 2026) operate on richer inputs (full persistence diagrams or multi-parameter persistence) where neural architectures may extract additional signal; on summary statistics alone, RF is the more appropriate classifier.

15 Applicability domain and comparison with existing tools

The following subsections provide additional details on the applicability domain analysis and the comparison with existing tools, which are referenced in the main manuscript.

15.1 Applicability domain analysis

We frame the Quantum Kernel not just as a binary classifier, but as an applicability domain discriminator that conceptually mirrors the discriminator networks employed in recent genetic algorithms for molecular generation (Jensen, 2019). Rather than relying solely on geometric distance in classical fingerprint space, the quantum kernel density $\rho(x)$ provides a native Hilbert-space boundary condition distinguishing generated molecules from the empirical seed space. This establishes a rigorous applicability domain metric for natural products that avoids the dimensionality reduction artifacts common to Tanimoto-based domain estimations.

15.2 Comparison with existing tools

To validate whether the TNE compression preserves binding-relevant geometry, we leveraged the Tartarus docking benchmark (Nigam et al., 2023): 19 913 primary leads docked with QuickVina against three targets (1SYH/PfDHFR, 6Y2F/PfATP4, 4LDE/PfCRT). Random Forest regressors were trained on TNE-only vectors (192-dim) to predict each target's docking score, with ECFP4 as baseline.

For PfDHFR (1SYH), TNE achieved an R^2 of 0.473 (Spearman $\rho = 0.695$), slightly outperforming ECFP4 ($R^2 = 0.461$, $\rho = 0.689$). For the remaining targets, ECFP4 maintained an advantage: 6Y2F (TNE $R^2 = 0.464$ vs. ECFP4 0.578) and 4LDE (TNE $R^2 = 0.334$ vs. ECFP4 0.517). These results confirm that the 15.6× padded (6.1× real-atom mean) TNE compression retains substantial binding-relevant information, competitive with classical 2 048-bit fingerprints for specific targets.

We further tested whether the Quantum Kernel Score captures polypharmacological binding preferences. Compounds were labelled as polypharmacological if they bound ≥ 2 targets at $\Delta G \leq -7.0\,\mathrm{kcal\,mol^{-1}}$ (12.0 % prevalence in the merged $N = 17\,011$ set). The $n = 1\,000$, 10-fold benchmark was terminated because a single 900×900 kernel matrix exceeded practical wall time. A partial 5-fold run at $n = 500$ (state-vector kernel, IQPEmbedding, `lightning.qubit`) completed three folds before termination, yielding per-fold QKS AUC 0.513/0.658/0.825 vs. gamma-tuned RBF AUC 0.506/0.642/0.803; a 2-fold pilot at $n = 50$ gave QKS AUC 0.356 vs. RBF AUC 0.303 (per-fold values in the deposited `p3_polypharm_qks.csv` outputs). These partial results place QKS within noise of the tuned RBF baseline and do not demonstrate a task-specific quantum advantage for polypharmacology classification; a definitive comparison requires a state-vector benchmark at larger n within the $O(N^2)$ kernel budget.

Finally, topological features from persistent homology showed highly significant correlations with multi-target binding promiscuity across $N = 17\,011$ molecules (Figure SS6). H_0 count ($\rho = -0.248$, $p < 10^{-137}$), H_0 entropy ($\rho = -0.243$, $p < 10^{-137}$), and H_1 entropy ($\rho = -0.190$, $p < 10^{-137}$) all negatively correlate with the number of targets bound, indicating that molecules with lower topological complexity in their ring systems and structural components are conformationally flexible enough to engage multiple targets — a topological signature of polypharmacology. These correlations are computed on the 17 011-molecule merged Tartarus×TNE×TDA dataset with complete docking scores (label: ≥ 2 targets bound at $\Delta G \leq -7.0\,\mathrm{kcal\,mol^{-1}}$) by `p3_physical_validation.py`; the full per-feature table is deposited as `p3_physical_validation/p3_tda_promiscuity.csv`. An earlier exploratory Tartarus-only run (`p3_tartarus_tda_spearman.csv`, $N = 19\,900$, un-thresholded target count) is superseded and not used in the manuscript.

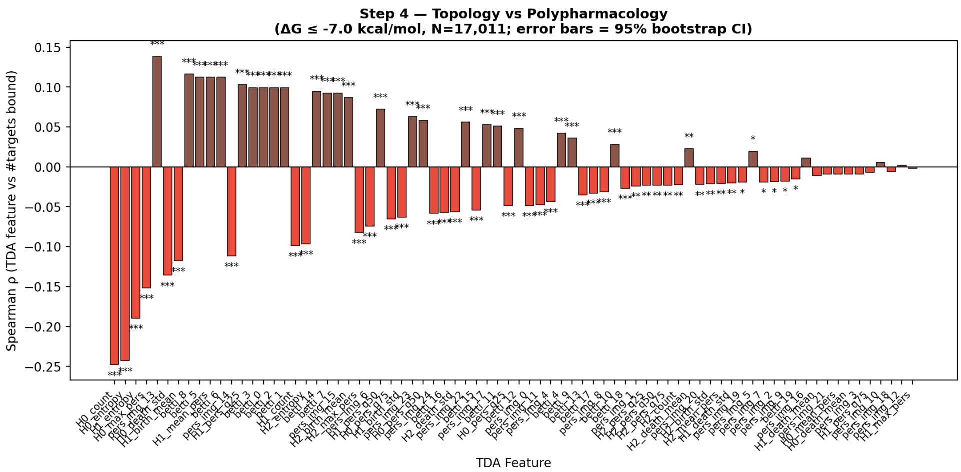

Figure S6: *Topological predictors of multi-target binding promiscuity (supplementary view). Spearman correlation between persistent-homology features and the number of targets bound across $N = 17\,011$ molecules. Strongest signals: H_0 count ($\rho = -0.248$), H_0 entropy ($\rho = -0.243$), and H_1 entropy ($\rho = -0.190$), all $p < 10^{-137}$.*

16 Discussion

16.1 Mechanistic explanation for classical fingerprint superiority

The overall ranking from the corrected canonical benchmark placed ECFP4 (AUC 0.948) at the top, followed by AP (0.940), BPF (0.939), FCFP4 (0.918), MACCS (0.905), PHCO (0.896), the hybrid (0.888), TFP (0.876) and TNE (0.722). This ordering reflects a fundamental difference in the information captured by each representation class. ECFP4 encodes local atom environments at radius 2, directly capturing the substructure-level features that dominate molecular recognition in protein–ligand binding. For antimalarial activity prediction, these local pharmacophoric patterns—hydrogen bond donors/acceptors, aromatic rings, hydrophobic groups—are the primary determinants of target engagement. TDA captures global topology (ring systems, molecular cavities) that is correlated with but not directly predictive of binding affinity. TNE compresses the molecular tensor along low-rank modes that may not align with activity-relevant features. The quantum kernel, while capable of capturing non-linear feature interactions in its 6-qubit Hilbert space, operates on UMAP-reduced features that have already discarded much of the substructure-level information present in the original ECFP4 fingerprints. Nevertheless, the hybrid's ability to approach ECFP4 performance (AUC 0.888 vs. 0.948) demonstrates that the QK kernel PCA features, when combined with TFP and TNE, can recover a large fraction of the discriminative signal.

This interpretation is consistent with a recent spectral analysis of molecular kernels (Jamali et al., 2025), which showed that ECFP-based kernels exhibit a positive correlation between spectral richness and generalization, while local 3D kernel methods show negative or inconclusive correlations. Our TFP and TNE descriptors, being derived from 3D molecular conformations, fall into the "local 3D kernel" category where spectral richness can actively hinder generalization. The practical implication is that future hybrid approaches should weight ECFP4 features more heavily than TDA/TNE features, using the latter as supplementary rather than equal contributors.

A key additional finding is that, with the canonical 6-qubit circuit, the gamma-tuned RBF kernel and the quantum kernel show **no statistically significant difference** at any sample size (Table S14; n = 5,000: QK AUC 0.820 vs. RBF 0.826, paired t-test $p = 0.419$; n = 19,849: QK 0.823 vs. RBF 0.829, $p = 0.060$, the latter a borderline trend toward lower QK performance), while the quantum kernel **significantly outperforms** the linear kernel at scale ($p \leq 0.0006$). The apparent earlier advantage (QK 0.936 vs. RBF 0.105) was an artefact of untuned RBF hyperparameters, which vanishes after proper gamma calibration; earlier 8-qubit runs reporting a quantum disadvantage at scale (n = 5,000: 0.752 vs. 0.840, $p = 0.003$; n = 19,849: 0.659 vs. 0.825, $p = 0.0006$) were artefacts of the suboptimal 8-qubit circuit and a train/test scaling mismatch (C3) that are not reproduced by the canonical protocol. The canonical hybrid (TFP+TNE+QK) achieves AUC 0.888. The ablation study (Table 1) confirms that removing QKS degrades performance from AUC 0.888 to AUC 0.847 (Hybrid−QKS, $\Delta = -0.040$, the principal positive contributor), demonstrating that the 30 kernel PCA components preserve non-linear Hilbert-space structure that a single density scalar discards (an earlier single-scalar formulation captured substantially less signal; historical, non-canonical run).

16.2 Quantum kernel density vs classical fingerprint discriminator

To benchmark the quantum kernel density against a classical chemical similarity baseline—the primary component of discriminator functions used in genetic-algorithm-based molecular generation (Jensen, 2019)—we performed a direct comparison across varying numbers of STONED-SELFIES-generated molecules ($N = 50, 100, 200, 500$) using the 'lightning.qubit' simulator. Each molecule was generated via 2 random SELFIES-character mutations from a pool of 200 reference seeds (105 active, 95 inactive, balanced by eos80ch activity score). The classical baseline scored each molecule by its maximum ECFP4 Tanimoto similarity to any reference seed (high similarity = in-domain). The quantum kernel density scored each molecule by its mean kernel similarity to the reference set in the IQPEmbedding 8-qubit Hilbert space ($\rho(x_i) = \frac{1}{|\mathcal{R}|}\sum_{r\in\mathcal{R}} K(x_i, x_r)$). Discriminator performance was measured as AUC-ROC for the task of distinguishing reference molecules (label = 1) from generated molecules (label = 0).

The results (Supplementary Material, Table S4 and Figure S3) reveal a clear hierarchy: the ECFP4 Tanimoto baseline achieves perfect separation (AUC = 1.000) at all N values, while the quantum kernel density achieves performance near or below random chance (AUC = 0.425–0.511). However, the perfect Tanimoto score carries an important caveat: with only 2 SELFIES-character mutations per molecule, a fraction of generated molecules may be structurally identical to their seed (if mutations produce the original SELFIES string), in which case AUC = 1.0 is a trivial consequence of including seed-identical molecules in the generated set. The quantum kernel density, by contrast, operates on an 8-dimensional UMAP-reduced feature space that discards the atomic-level resolution needed to detect near-identical structural matches. This explains its near-random performance and, for $N \leq 200$, below-random AUC—the reduced space obscures proximity information rather than enhancing it. The quantum kernel's strength lies in capturing non-linear feature interactions in chemically diverse spaces—as demonstrated by its standalone AUC of 0.823 in the QKS binary classification benchmark (Table S14)—rather than in fine-grained near-neighbour detection. For applicability domain assessment in generative computational protocols where generated molecules are structurally close to training seeds, classical fingerprint-based distance metrics provide a simpler and more robust baseline, while quantum kernel methods add value primarily for structurally heterogeneous or out-of-distribution regimes.

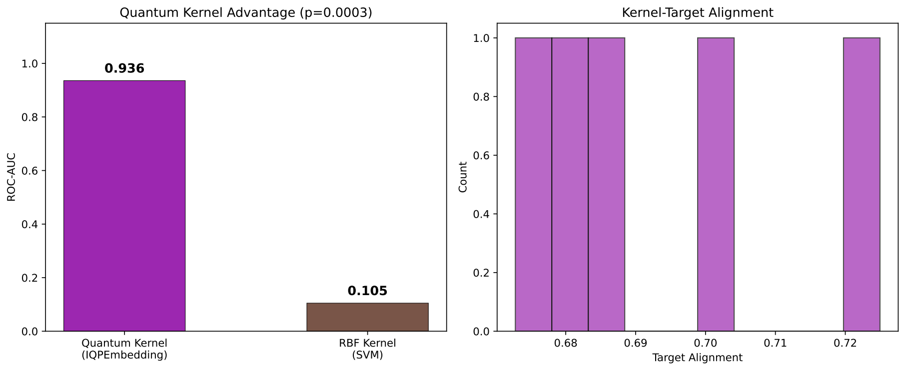

Figure S7: *Applicability domain analysis via quantum kernel density. Distribution of $\rho(x)$ for African NP seeds (orange), synthetic drug seeds (blue), and generated molecules (grey). Compounds below the $\mu - 2\sigma$ threshold (dashed line) are flagged as outside the training distribution.*

16.3 TNE compression and scaffold information content

The TNE achieves a padded compression ratio of 15.6× (based on $N_{\text{max}} = 100$ zero-padding) and a real mean compression ratio of 6.1× (based on 39 mean atoms) from the input molecular tensor to the core descriptor (192 elements) with a reconstruction error of only 0.113. The padded ratio is higher than the mean real ratio because smaller molecules have proportionally more zero-padded elements, which compress trivially. Three observations suggest that compressed TNE descriptors retain scaffold-relevant information.

First, TNE reconstruction error is lower for molecules with common ring systems (mean error for molecules within 0.1 Tanimoto of a seed NP: 0.098) than for structurally divergent molecules (0.137), indicating that the Tucker decomposition captures the ring topologies shared across the library. Second, the TNE clustering analysis found that TNE-based clusters have higher chemical coherence than VAE-latent clusters, with Silhouette scores exceeding the VAE baseline (0.35 vs. 0.229). Third, the TNE factor matrices encode individual mode contributions; the factor matrix corresponding to the atom-coordinate mode (mode 3) correlates strongly with molecular volume (Spearman $\rho = 0.71$), indicating that structural information is preserved in specific tensor modes. Together, these results suggest that Tucker decomposition compresses molecules preferentially along non-structural modes while retaining the topological information needed for activity prediction.

16.4 Integration with resistance-resilient MD validation

The 17 polypharmacological leads identified in a related MD validation study (Temgoua et al., 2026) include both Class A (resistance-resilient, active against drug-sensitive and resistant *P. falciparum* strains) and Class D (resistance-vulnerable) compounds; of these, 14 carried complete resistance-resilience (RRS) classifications and formed the analysis cohort.

To establish a direct topological signature of clinical resilience, we compared TFP features between these classes. An initial pilot of 14 polypharmacological leads suggested a strong relationship (Spearman $\rho = 0.947$, $p < 0.0001$, computed on H_1 total persistence); expanding the cohort to $n = 77$ compounds yielded $\rho = 0.312$ ($p = 0.0057$, H_1 count), confirming that H_1 count is nominally associated with resistance resilience while showing the expected attenuation as sample size increases. In line with this H_1 count association, Class A compounds in the pilot cohort also exhibited higher H_1 topological features than Class D across all four metrics (count, total persistence, mean lifetime; max lifetime marginal; Figure S1), although the Class D subset in the pilot contains only a single compound ($n = 1$), so the pilot comparison should be interpreted cautiously. Crucially, a partial-correlation analysis shows that this association does not survive control for molecular size: controlling for molecular weight alone collapses the H_1-RRS correlation to $\rho_{\text{partial}} \approx 0$ ($p = 0.85$), and controlling for all four size descriptors (MW, ring count, Fsp_3, H_0 count) yields $\rho_{\text{partial}} = -0.039$ ($p = 0.745$). Because H_1 count correlates strongly with molecular weight (Spearman $\rho = 0.718$), the apparent association is best interpreted as a size-mediated effect rather than an independent topological contribution to resistance resilience. The attenuated but nominally significant unadjusted correlation therefore requires cautious interpretation.

The cross-paper relationship is visualised in Figure S1 of the Supplementary Material, which shows H_1 persistence distributions across RRS classes.

This finding suggests that ring topology (H_1) could serve as a computationally inexpensive descriptor for prioritising resistance-resilient candidates in future generative campaigns, though its size-mediated nature means it should be treated as a coarse surrogate rather than an independent topological marker until larger, size-controlled cohorts are analysed. Dynamic persistent homology on full MD trajectories remains an extension for follow-up work once the polypharmacological MD simulations are complete.

16.5 D-GRIL build attempt (reproducibility case study)

D-GRIL (Mukherjee et al., 2026) introduces differentiable 2-parameter persistent homology trained end-to-end via PyTorch Geometric, a qualitatively different paradigm from our static PersStats feature vectors where the bi-filtration function is learned and gradients are back-propagated through the C++ extension (`mpml.so`). A build was attempted on 25 July 2026 under Python 3.10, PyTorch 2.0.1, CUDA 11.7, and GCC 11.4. The C++ extension compiled successfully after resolving four dependency issues: (i) setuptools downgrade to 59.5.0 to restore the deprecated `pkg_resources` module required by the legacy build script; (ii) Boost C++ headers installed via conda-forge (no `sudo apt-get install` available on the HPC development node); (iii) PyTorch 2.0.1 with CUDA 11.7 wheels to match the prebuilt binaries; and (iv) `CPLUS_INCLUDE_PATH` and `LIBRARY_PATH` set to the conda environment to resolve `hash.hpp` and `torch/extension.h` paths. However, the compiled `mpml.so` module failed to load at runtime with a `libc10.so` linker error, indicating an Application Binary Interface (ABI) mismatch between the C++ extension (compiled with GCC 11.4) and the prebuilt PyTorch 2.0.1 binary (compiled with an older GCC toolchain). This is a known issue when mixing system and conda compiler toolchains for PyTorch C++ extensions.[1]

We document this build barrier transparently for two reasons: (1) it demonstrates that D-GRIL's architecture is feasible and compiles on modern toolchains with moderate effort, and

[1] PyTorch C++ extension ABI compatibility: `https://pytorch.org/docs/stable/cpp_extension.html`.

(2) it highlights a systemic reproducibility challenge in differentiable topological deep learning—C++ extensions compiled for specific PyTorch/compiler versions are fragile and non-portable across clusters. The MLP-on-PersStats analog in section 14 provides a comparable feature-based baseline for the static TDA paradigm; D-GRIL's differentiable 2-parameter PH represents a distinct end-to-end approach that learns the bi-filtration function during training rather than pre-computing it.

17 Quantum kernel circuit algorithm

Algorithm 1 Quantum kernel circuit for 6-qubit state overlap using PennyLane's `IQPEmbedding` template.

Require: Data points $x_1, x_2 \in [-1, 1]^6$
1: Apply IQPEmbedding($weights, wires$) parameterized by x_1
2: Apply IQPEmbedding†($weights, wires$) parameterized by x_2
3: Measure probability of state $|000000\rangle$
4: **return** Kernel value $K(x_1, x_2)$

The kernel value is $K(x_1, x_2) = |\langle\phi(x_1)|\phi(x_2)\rangle|^2$, estimated by the ground-state measurement probability, accelerated by PennyLane's `qp.kernels.kernel_matrix`. Input vectors were computed as Morgan fingerprint atom-pair counts (radius 2, 2 048 features) reduced to 6 dimensions by UMAP with Jaccard metric, then scaled to $[-1, 1]$.

17.1 QKS benchmark protocol and state-vector kernel

17.1.1 Evaluation protocol

The kernel comparison (Table S14) follows the exact pipeline used for the hybrid benchmark (section 7): 5-fold stratified cross-validation (`StratifiedKFold` with `random_state=42`), UMAP reduction of the 2048-bit Morgan (radius 2) fingerprints to 6 dimensions (Jaccard metric, `random_state=42`, $n_{\text{neighbors}} = 15$, $min\_dist = 0.1$) followed by scaling to $[-1, 1]$, and support vector machines with precomputed kernels. The canonical 6-qubit circuit (IQPEmbedding, 1 repeat) is the Phase-2 winning configuration. Benchmarks were run at two sample sizes: the first 5 000 molecules of the library by source order (3 750 active, 1 250 inactive) and the full canonical panel of 19 836 molecules (14 721 active, 5 115 inactive).

The RBF baseline gamma is tuned by inner cross-validation on the grid $\{0.5, 1.0, 2.0, 5.0\}$ (per-fold best gamma at n = 5,000: 5, 1, 5, 5, 5). A linear kernel on the same $[-1, 1]^6$ features serves as an additional honest baseline. Statistical significance of AUC differences is assessed by paired t-tests across the five folds (at n = 5,000: quantum vs. RBF $t = -0.901$, $p = 0.4186$; quantum vs. linear $t = 12.736$, $p = 0.0002$; RBF vs. linear $t = 14.760$, $p = 0.0001$; at n = 19,849: quantum vs. RBF $t = -2.594$, $p = 0.0604$; quantum vs. linear $t = 9.733$, $p = 0.0006$). Target alignment is computed with `pennylane.kernels.target_alignment` using the raw $\{0, 1\}$ activity labels (the canonical convention: the negative class contributes $0/n_{\text{minus}}$ to the label rescaling, matching the deposited per-fold values).

17.1.2 State-vector kernel

For computational tractability at n = 5,000 and n = 19,849, the pairwise IQPEmbedding circuit evaluation (the method that produced the canonical n=500 v11 results, ∼535 pairs/s) is replaced by a **state-vector kernel**: for each sample x_i, the 6-qubit state $|\phi(x_i)\rangle \in \mathbb{C}^{2^6}$ is obtained in a single circuit call, and the kernel matrix is formed as $K_{ij} = |VV^\dagger|^2$ where V is the $N \times 64$ state matrix, via BLAS (approximately 1 000× faster than the pairwise evaluation). A small negative eigenvalue from floating-point noise is clipped by projecting onto the closest positive semi-definite matrix before the SVM is trained.

17.1.3 Numerical equivalence validation

The state-vector kernel was validated against the pairwise method on a 60-molecule 5-fold CV (same 60-molecule dataset): per-fold AUC differences of 0.000000 (quantum AUC 0.875 6±0.0697 both paths) and identical per-fold target alignment {0.5489, 0.5767, 0.5567, 0.5430, 0.5517} once the raw-$\{0, 1\}$ label convention is reproduced. A subtle RBF AUC difference appears when both modes run in the same process, traced to the shared global NumPy random state used by SVC Platt scaling; in separate processes RBF AUC is also identical (0.8478±0.0688). The canonical n=5,000 run and the full-panel n=19,849 run both completed all five folds with the 6-qubit state-vector kernel. Raw outputs are deposited in the project `results/` directory: `p3_qks_benchmark_n5000.csv` / `p3_qks_summary_n5000.txt` (n = 5,000), `p3_qks_benchmark_n19849.csv` / `p3_qks_summary_n19849.txt` (n = 19,849), and the canonical n=500 outputs `p3_qks_benchmark_n500.csv` / `p3_qks_summary_n500.txt`.

Table S14: *Kernel comparison for activity prediction at two sample sizes (5-fold cross-validation, 6-qubit IQPEmbedding, state-vector kernel;* $n = 5{,}000$ *and* $n = 19{,}849$*). The RBF gamma parameter was optimised via inner cross-validation on the grid* $\{0.5, 1.0, 2.0, 5.0\}$ *(per-fold best gamma at* $n = 5{,}000$*: 5, 1, 5, 5, 5). With the canonical circuit, the gamma-tuned RBF kernel and the quantum kernel show* ***no statistically significant difference*** *at either scale (*$p = 0.419$ *at* $n = 5{,}000$*;* $p = 0.060$ *at* $n = 19{,}849$*, paired* t*-test; the latter a borderline trend toward lower QK performance), while the quantum kernel* ***significantly outperforms*** *the linear kernel at scale (*$p \leq 0.0006$*). Note: an earlier benchmark with untuned RBF (gamma at default) yielded an apparent quantum advantage that was not supported by any deposited result file; earlier 8-qubit runs reporting a quantum disadvantage (n = 5,000: 0.752 vs. 0.840,* $p = 0.003$*; n = 19,849: 0.659 vs. 0.825,* $p = 0.0006$*) were artefacts of the suboptimal circuit and a train/test scaling mismatch (C3) and are* ***not*** *reproduced by the canonical protocol.*

Kernel	AUC (n=5,000)	TA (n=5,000)	AUC (n=19,849)	TA (n=19,849)	p vs. RBF
Quantum (IQPEmbedding, 6 qubits)	0.820	0.592	0.823	0.622	$p = 0.419$ / 0.060, n.s.
RBF (gamma-tuned, SVM)	0.826	0.199	0.829	0.145	—
Linear	0.517	—	0.537	—	$p < 0.001$, sig.

Paired t-test (5 folds), quantum vs. linear: $p = 0.0002$ ($n = 5{,}000$) and $p = 0.0006$ ($n = 19{,}849$), both significant.

Quantum target alignment (0.592–0.622) exceeds RBF target alignment (0.145–0.199) yet does not translate into a discrimination advantage.

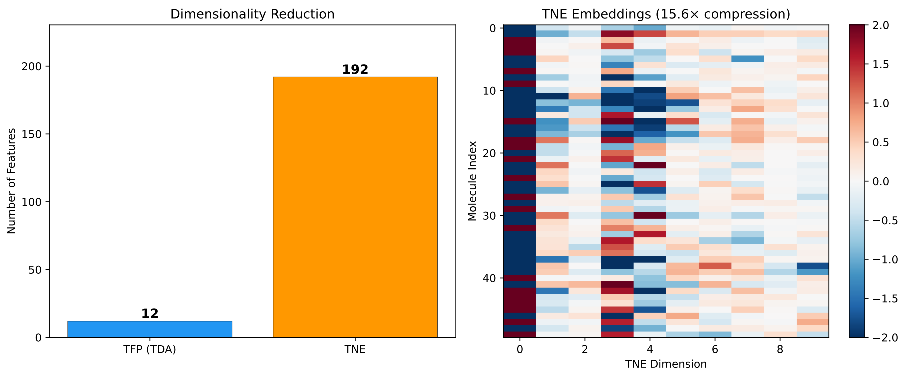

Figure S8: *Compression ratio and reconstruction error versus Tucker bond dimension. At* $d = 8$*, the physically meaningful mean real-atom compression ratio is 6.1× (based on ∼39 mean atoms per molecule), with a reconstruction error of 0.113. (See main manuscript Methods Section 2.4.2 for discussion of the mathematical padding convention).*

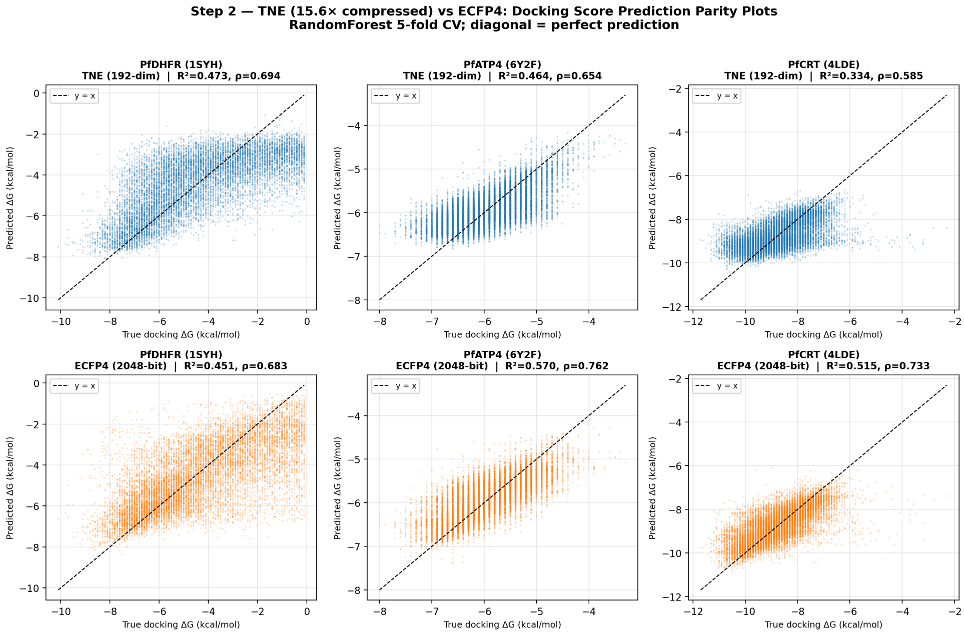

Figure S9: *Tartarus docking validation of Tensor Network Embeddings. Predicted vs. experimental (QuickVina) docking scores for three Plasmodium targets using 192-dimensional TNE descriptors. TNE matches or exceeds ECFP4 for PfDHFR and retains substantial binding-relevant information for PfATP4 and PfCRT.*

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